



What Explains a Pitcher's Value?

Spin Rate, Velocity and Movement as predictors of WAR in the MLB

BY: Shannon Maccallum

For this project I wanted to answer the question “How do a pitcher’s spin rate, velocity, and pitch movement relate to their on-field value as measured by WAR?” I wanted to explore this, by not only looking at one player, or one season, but as a combination of many players over the course of 5 seasons. WAR (Wins Above Replacement) was the metric that I decided to dive into in order to measure how valuable a pitcher in the MLB is.

The three pitching variables for 4 seam fastballs I chose to look at were Spin Rate, Velocity, and Movement (horizontal and vertical). I obtained data for MLB pitchers from 2021 to 2025 using Statcast, a public baseball statistic website. I filtered down to the 93 pitchers who had complete data across all 5 seasons, ensuring consistent comparison. The variable, WAR, was collected from yearly MLB data on the public data on Baseball reference for all pitchers and averaged over 5 years, for the 93 players.

Each pitcher's spin rate, velocity, and movement values were averaged across 5 seasons. Movement was broken into horizontal and vertical components and also analyzed as a total. Pitchers were then classified into high/low groups (H/L) using the median of each metric. This created 8 profiles (e.g., HHH, HHL, LLL), that could then be used to explore the effects of different variables. The tools that were used included Excel, to clean data and extract from the web, and R used for data cleaning, merging, regression modeling, visualizations.

After gathering and using R to produce the data needed, I looked at multiple forms of analysis to understand the relationships between the variables. Visualizations suggested that velocity and vertical movement might positively relate to WAR. In scatter plots, pitchers with higher average fastball velocity and greater vertical break tended to cluster with higher WAR values. Horizontal movement and spin rate showed weaker trends or more scattered patterns.

The scatterplot of average 5-year velocity vs WAR, colored by profile group, reveals that no single profile dominates the upper-right quadrant. However, the trends that emerged give insight into the relationships of the variables and WAR. For example, LHL and LLL, surprisingly, include many of the highest WAR performers despite having low velocity. On the other hand, several HHH and HHL pitchers cluster below the WAR = 2 line, indicating that velocity alone does not guarantee greater value. The results showed that there were meaningful differences in WAR across profile groups, which suggests that combinations of pitch traits may matter more than individual metrics alone. The data gave us insights that at the MLB level, other factors like command, deception, pitch sequencing, or role (starter vs reliever) likely matter more than a change in velocity, spin rate and movement.

The distribution of average spin rate across the 93 pitchers is approximately normal with a slight left skew. The mean spin rate is about 88.4 RPM, with a median of 88.0 RPM. The standard deviation is roughly 5.3 RPM, with a 95% margin of error of approximately ± 1.1 RPM. This shows there is moderate variability among pitchers. The correlation between spin rate and WAR is very weak ($r \approx -0.03$) and not statistically significant ($p = 0.716$). This suggests that spin rate alone does not meaningfully predict a pitcher's on-field value, as measured by WAR.

The distribution of average fastball velocity is approximately normal, with a slight right skew. The mean velocity across the sample is 93.7 mph, and the median is 93.5 mph. The

standard deviation is around 2.1 mph with a 95% margin of error of approximately ± 0.4 mph. The correlation between velocity and WAR is ~~weakly~~ positive ($r \approx 0.10$), but not statistically significant ($p = 0.905$). Although higher-velocity pitchers tend to visually cluster at higher WAR levels, velocity alone is not a reliable predictor of pitcher value in this dataset.

When analyzing pitch movement, vertical and horizontal components were considered separately. Vertical movement has a roughly normal distribution, while horizontal movement shows greater variability. Combined, the total movement distribution has a mean of 17.2 inches and a standard deviation of about 3.4 inches. The 95% margin of error is around ± 0.7 inches. Correlation tests showed that vertical movement ($r \approx -0.05$) and horizontal movement ($r \approx -0.08$) both had negligible relationships with WAR, and neither result was statistically significant ($p > 0.60$). These findings suggest that while movement may contribute to pitch quality, it does not independently predict a pitcher's WAR when looked at alone.

At the MLB level, the pitchers are so advanced that their own individual pitching style likely affects their WAR far more than any technical skills. ~~I would be very interested to look at this project from another level, college or even high school.~~



